Predict a position and a range

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict a future position and explain the limits of the prediction.

SUCCESS CRITERIA - I can...

1. Use matching units and include the starting position.
2. Report a central prediction and its bounds.
3. Distinguish a measurement range from model limitations.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “initial” and “final” to explain your model or result. Sentence starters are optional.

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

bound: an upper or lower limit in a model

01 Warm-up: bring back an earlier idea

Start at 5 m, move to 14 m, then finish at 7 m.
Draw the route.
Find distance, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $x_f = x_i + v\,\Delta t$.
Constant velocity; i = initial, f = final. Use the same units for all positions.

Worked reasoning: A cart starts at 1 m and moves right at 2 m/s for 3 s. Central prediction: 1+2×3=7 m. If its speed could be 1 to 3 m/s, the endpoint range is 4 to 10 m.

Your turn: A cart moves along a marked track.
Show its start, central predicted endpoint, and full possible endpoint range on the number line. What must remain constant throughout the 3 s?

Sentence starter: I predict ... . My model supports this because ... .



Position (m); right is positive

Marzano check - circle: ① ② ③

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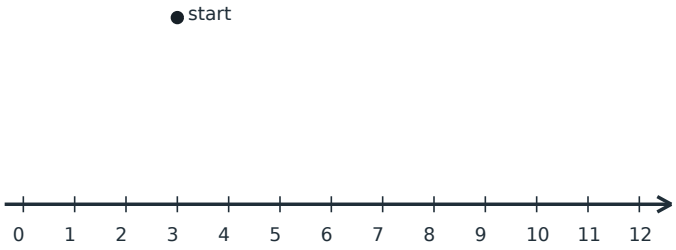
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Start plus change

A cart starts at 3 m and moves right at 2 m/s for 4 s.
Find its displacement and final position.
Mark the endpoint.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Predict leftward motion

A robot starts at the 12 m mark and drives left at a steady 2 m/s for 3 s.
Draw the move on the number line.
Find the displacement and final position.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Position (m); right is positive

Marzano check - circle: ① ② ③

05 Match the time units

A rover moves at 2 m/s for 1 minute.
Use 1 minute = 60 seconds.
Find its distance at this steady speed.

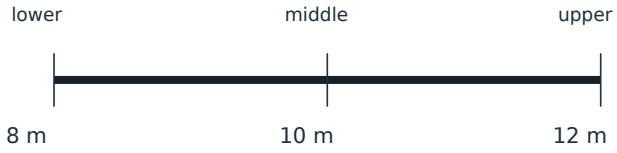
Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

06 Read the range

A prediction is 10 m with a range from 8 m to 12 m.
Label the middle, lower end, and upper end.
Find how far each end is from the middle.

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

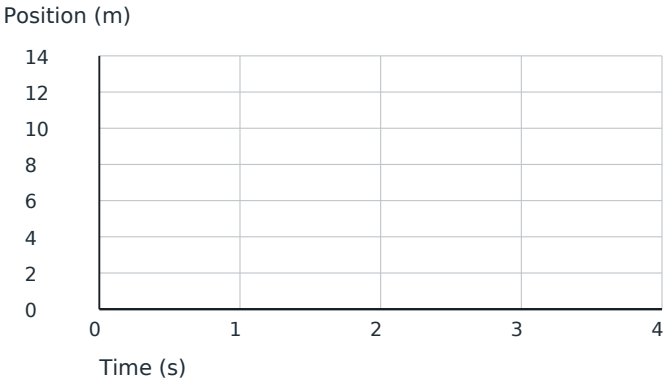
Sentence starters are optional. You may plan in any language.

07 A slow and a fast case

A rover starts at 0 m and travels right at a constant speed somewhere between 1 and 3 m/s.
Draw the slowest and fastest possible position-time lines.
Find the possible position range after 4 s.

Support: Slow case: ____. Fast case: ____. My range is ____.

Sentence starter: I found ... [unit]. This value means ... in this situation.



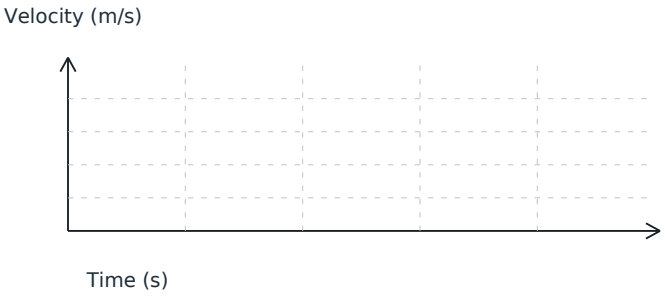
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A velocity graph is constant at -3 m/s for 5 s.
Find displacement. A second object has components 7 m/s east and 2 m/s south; sketch its direction.
Sketch the velocity-time model; label the given values.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

09 Exit check

A cart starts at 2 m and moves right at 3 m/s for 2 s.
Find the central final position. If the possible final positions run from 7 to 9 m, state the range and one assumption.

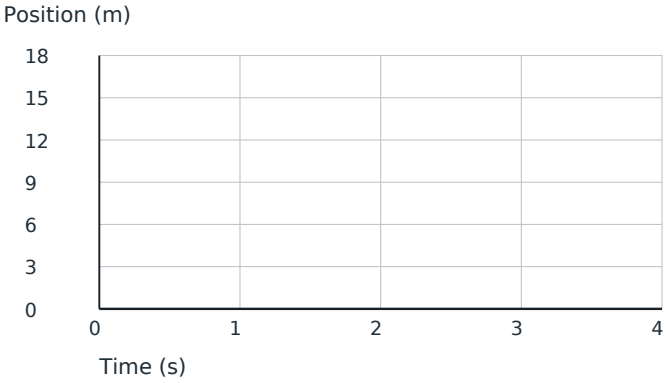
Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

10 STRETCH: test the model

Design two different starting-position and speed pairs that both give an endpoint of 10 m after 2 s.
Draw the routes.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3